# From Words to Worlds

Fine-tuning Stable Diffusion with LoRA, and BERT vs. CLIP through a trained connector

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## 1. Summary

Every other run in the group uses Stable Diffusion 1.5 frozen. This report covers the two runs where weights are actually trained. **Run 1** fine-tunes the UNet with LoRA adapters on COCO and measures whether it follows prompts better. **Run 2** replaces SD's text reader: CLIP and BERT each go through an identical, newly trained connector, with and without LoRA, to test how much the choice of text encoder matters once both have to be learned through the same bridge. Both runs use the same 3,000 COCO train2017 pairs for training and the same 40 held-out val2017 captions and seeds for scoring, so their numbers are directly comparable.

| Result | Value | What it shows |
|---|---|---|
| LoRA, trainable share of the UNet | 1.59 M of 861 M (0.185%) | A very small add-on; 13.8 min on a free T4, 2.6 GB GPU memory |
| LoRA, CLIPScore vs base | 25.52 → 26.37 (+0.84, 95% CI 0.16 to 1.53) | A small but real improvement in prompt-following |
| LoRA, win rate vs base | 72% of 40 prompts | The gain is spread across prompts, not one outlier |
| LoRA, held-out validation loss | 0.15120 → 0.15102 | Denoising barely changed; no sign of overfitting |
| Connector, BERT vs CLIP (connector only) | −1.95 (95% CI −3.47 to −0.50) | Through the same connector, BERT is worse than CLIP |
| Connector, BERT vs CLIP (+ LoRA) | −2.16 (95% CI −3.96 to −0.41) | Same result with LoRA added |
| Connector runs vs base | −11 to −14 CLIPScore | 600 steps cannot train a connector from scratch; all four runs lose most of the prompt |
| Does LoRA help a connector? | CIs include 0 for both | No measurable effect at this training length |

## 2. Why this question, and why it is independent of the other runs

Santhosh's work studies the frozen model's attention (when each word is painted), Tanmayi's builds a cleaned training set, and Praniti's builds evaluation sets and compares text encoders without training. None of them updates any weights. This work asks the training question directly: *how much of Stable Diffusion needs to be trained to change what it draws, and what happens when the text side has to be learned too?* It also tests, with generated images, the prediction in Praniti's report that BERT starts behind CLIP (CKA 0.53).

## 3. Setup and design choices

| Choice | Why |
|---|---|
| Train on COCO **train2017**, evaluate on **val2017** | The group's frozen evaluation splits come from val2017, so training never sees an evaluation image. |
| 3,000 images, 15,004 captions; one caption sampled per step | Inside the milestone's 1k–5k range. Keeping all captions of an image acts as free text augmentation. |
| Eligibility: short side ≥ 384 px, aspect ≤ 2, ≥ 3 clean captions | 97,799 train and 4,072 val images qualified. Small or very wide photos would be blurred or heavily cropped at 512 × 512. |
| Center crop to 512 × 512; VAE latents cached once, plus a flipped copy | SD 1.5's native size. Caching means each step only runs the UNet; flips double the images without new downloads. |
| LoRA rank 8 on every UNet attention projection (to_q, to_k, to_v, to_out) | Trains 0.185% of the UNet. Full fine-tuning (≈14 GB of weights, gradients and optimizer state) does not fit a free T4 and overfits 3k images. |
| Standard objective: predict the added noise (MSE) | The same loss SD was trained with, so any change comes from the data, not a new objective. |
| AdamW 1e-4, 50-step warmup + cosine, effective batch 4, 600 steps | Conservative LoRA settings. 600 steps (2,400 images, under one pass) keeps the pilot to about 14 minutes. |
| 10% of captions replaced with an empty string | Keeps the unconditional branch that classifier-free guidance needs at sampling time. |
| fp16 autocast + GradScaler, gradient checkpointing; trainable weights kept in fp32 | Fits a T4 comfortably (peak 2.6 GB) without the precision loss of fp16 updates. |

## 4. How it is evaluated

**Validation loss**: the training objective on 100 held-out val2017 images, with the noise and timesteps fixed so every checkpoint is scored on exactly the same problem. **CLIPScore** (100 × cosine between image and caption, CLIP ViT-L/14, the same CLIP the rest of the group uses) on 40 held-out val2017 captions. The base model and every checkpoint generate from the **same prompts and seeds** (DPM-Solver, 25 steps, guidance 7.5), so each comparison is paired, and a **paired bootstrap** (5,000 resamples) gives the 95% interval. Images blacked out by the safety checker are excluded pairwise (none were in this run). Five more held-out captions give the showcase images.

## 5. Results

| Checkpoint | CLIPScore | Δ vs base [95% CI] | Beats base on | Validation loss |
|---|---|---|---|---|
| base | 25.52 | — | — | 0.15120 |
| step 150 | 25.96 | +0.44 [−0.11, +1.01] | 50% | 0.15107 |
| step 300 | 26.28 | +0.75 [+0.14, +1.38] | 62% | 0.15106 |
| step 450 | 26.29 | +0.77 [+0.16, +1.40] | 65% | 0.15103 |
| step 600 | **26.37** | **+0.84 [+0.16, +1.53]** | **72%** | 0.15102 |

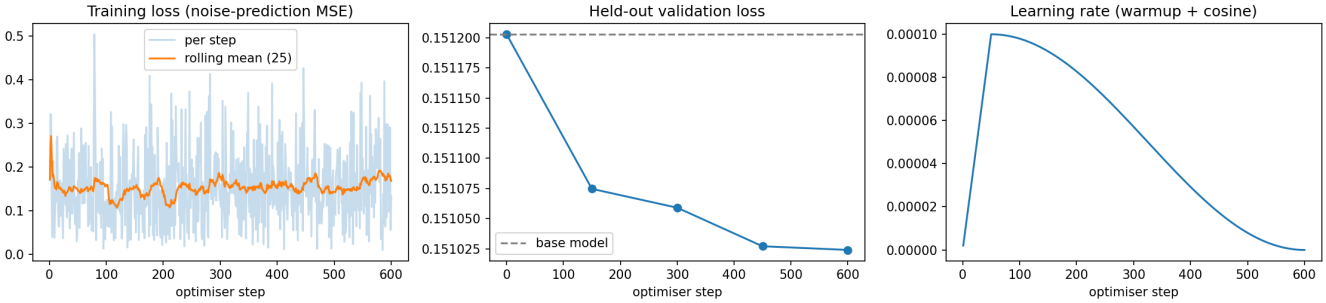



*Figure 1. Training loss per step (noisy because each step draws a random timestep), held-out validation loss at each checkpoint, and the learning-rate schedule.*

**Observations.** (1) From step 300 onward the interval stays above zero, so the gain is not noise, and by step 600 the fine-tuned model beats the base on 72% of prompts. (2) Most of the gain arrives by step 300; 300 → 450 adds only 0.01, and 450 → 600 adds 0.08, so it is slowing rather than plateaued. (3) Validation loss falls at every checkpoint but only by 0.1% in total: LoRA changes how the UNet uses the text far more than how well it denoises, and there is no sign of overfitting. (4) Training loss is not informative on its own: its variance comes almost entirely from the random timestep.

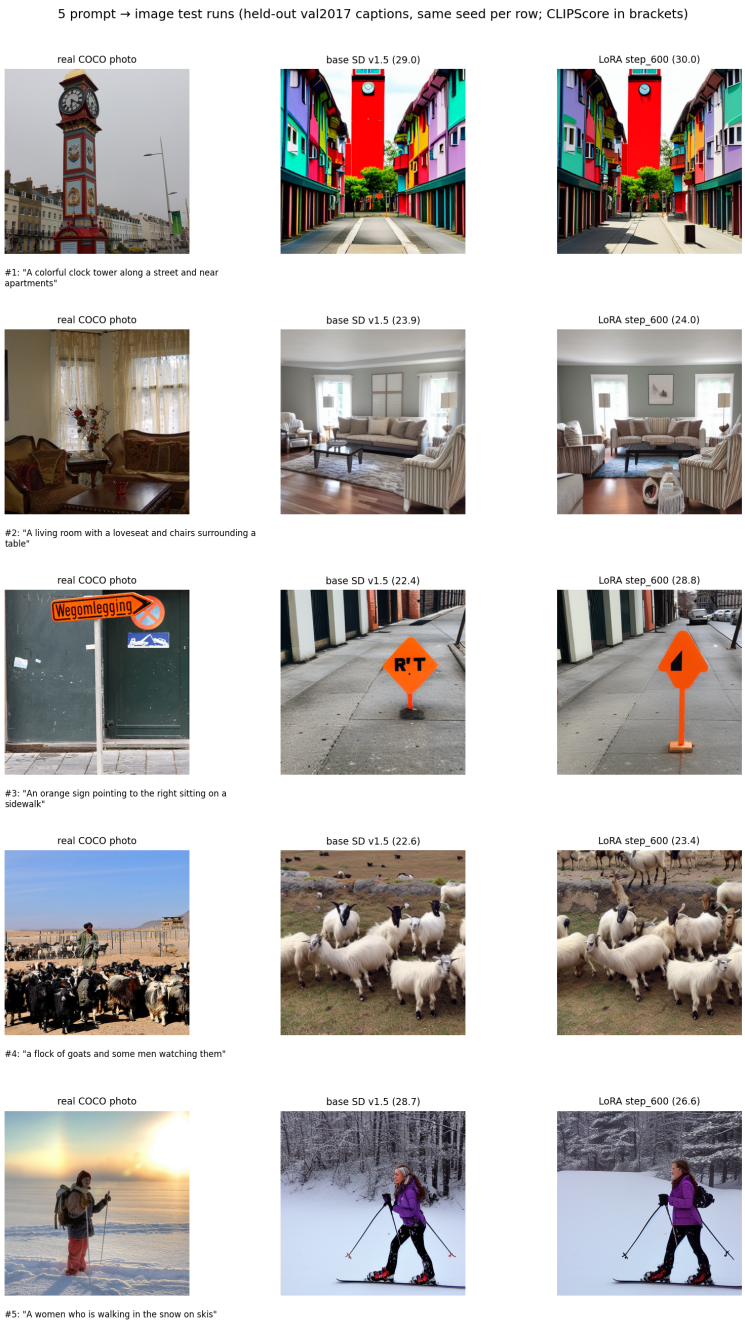



*Figure 2. The five showcase prompts: real COCO photo, base SD 1.5, and LoRA after 600 steps, same seed per row (CLIPScore in brackets).*

| Prompt | What the samples show |
|---|---|
| A colorful clock tower along a street and near apartments | Same composition; LoRA keeps the scene and edges up (29.0 → 30.0). |
| A living room with a loveseat and chairs surrounding a table | Near-identical room with small furniture changes (23.9 → 24.0). |
| An orange sign pointing to the right sitting on a sidewalk | **Clearest win.** Base draws a diamond sign with garbled text; LoRA draws an arrow-shaped sign (22.4 → 28.8). |
| a flock of goats and some men watching them | Both draw the goats and **neither draws the men**: relations between objects remain hard (22.6 → 23.4). |
| A women who is walking in the snow on skis | **Slightly worse** (28.7 → 26.6); the skier is similar but the scene is plainer. |




*Figure 3. One prompt across checkpoints. With a fixed seed the layout is set by the base model; LoRA changes details, not composition.*

## 6. Question and design

SD 1.5's UNet only ever learned to read CLIP's text features. The group's proposal plans to replace CLIP with BERT through a trained connector. A fair test cannot compare *CLIP used directly* with *BERT through a freshly trained connector*, because that mixes up two changes. So both encoders go through **the same connector, trained the same way**, and CLIP-through-a-connector is the control.

| Run | Text encoder (frozen) | Trained | Question it answers |
|---|---|---|---|
| base | CLIP, direct | nothing | Reference |
| clip_conn | CLIP | connector | What does a from-scratch connector cost? (control) |
| bert_conn | BERT-base | connector | BERT vs CLIP, same connector |
| clip_conn_lora | CLIP | connector + UNet LoRA | Does LoRA help the UNet absorb the connector? |
| bert_conn_lora | BERT-base | connector + UNet LoRA | Does LoRA help the UNet adapt to BERT? |

| Choice | Why |
|---|---|
| Connector: LayerNorm → Linear(768→1024) → GELU → Linear(1024→768) → LayerNorm, random init, 1.58 M params | About the same size as the rank-8 LoRA (1.59 M). The output LayerNorm matches CLIP, whose last hidden state is itself a LayerNorm output. BERT-base and CLIP ViT-L/14 are both 768 wide, so the shape is identical for both. |
| Same random initialisation for both connectors (no residual path) | A residual connection would hand CLIP its original features for free and BERT nothing; random init makes both start equally far from what the UNet expects. |
| Token states padded/truncated to 77, no attention mask | Exactly how SD 1.5 feeds CLIP, applied to both encoders. |
| Connector lr 5e-4, LoRA lr 1e-4; everything else as Part 1 | The connector starts from scratch so it needs a larger step; data, steps, batch order (same seed), prompts and seeds are identical across runs. |
| A fresh UNet for every run | No run inherits another's training. |

## 7. Results

| Run | Trainable (M) | CLIPScore | Δ vs base [95% CI] | Beats base on | Validation loss |
|---|---|---|---|---|---|
| base (CLIP direct) | 0 | 25.50 | — | — | 0.15120 |
| clip_conn | 1.58 | 14.35 | −11.20 [−13.11, −9.38] | 3% | 0.15241 |
| bert_conn | 1.58 | 12.43 | −13.07 [−14.71, −11.48] | 0% | 0.15238 |
| clip_conn_lora | 3.17 | 13.69 | −11.81 [−13.65, −10.01] | 3% | 0.15197 |
| bert_conn_lora | 3.17 | 11.83 | −13.73 [−15.16, −12.28] | 0% | 0.15202 |

| Question | Comparison | Δ CLIPScore [95% CI] | Answer |
|---|---|---|---|
| BERT vs CLIP, connector only | bert_conn − clip_conn | −1.95 [−3.47, −0.50] | **BERT worse** |
| BERT vs CLIP, connector + LoRA | bert_conn_lora − clip_conn_lora | −2.16 [−3.96, −0.41] | **BERT worse** |
| Does LoRA help BERT? | bert_conn_lora − bert_conn | −0.81 [−2.13, +0.49] | No measurable effect |
| Does LoRA help CLIP + connector? | clip_conn_lora − clip_conn | −0.51 [−2.30, +1.35] | No measurable effect |

The base model scored 25.50 here and 25.52 in Part 1: the same prompts and seeds in a separate session, so the two runs line up. The safety checker blacked out at least one image in the connector + LoRA runs; flagged images are excluded pairwise from every comparison.

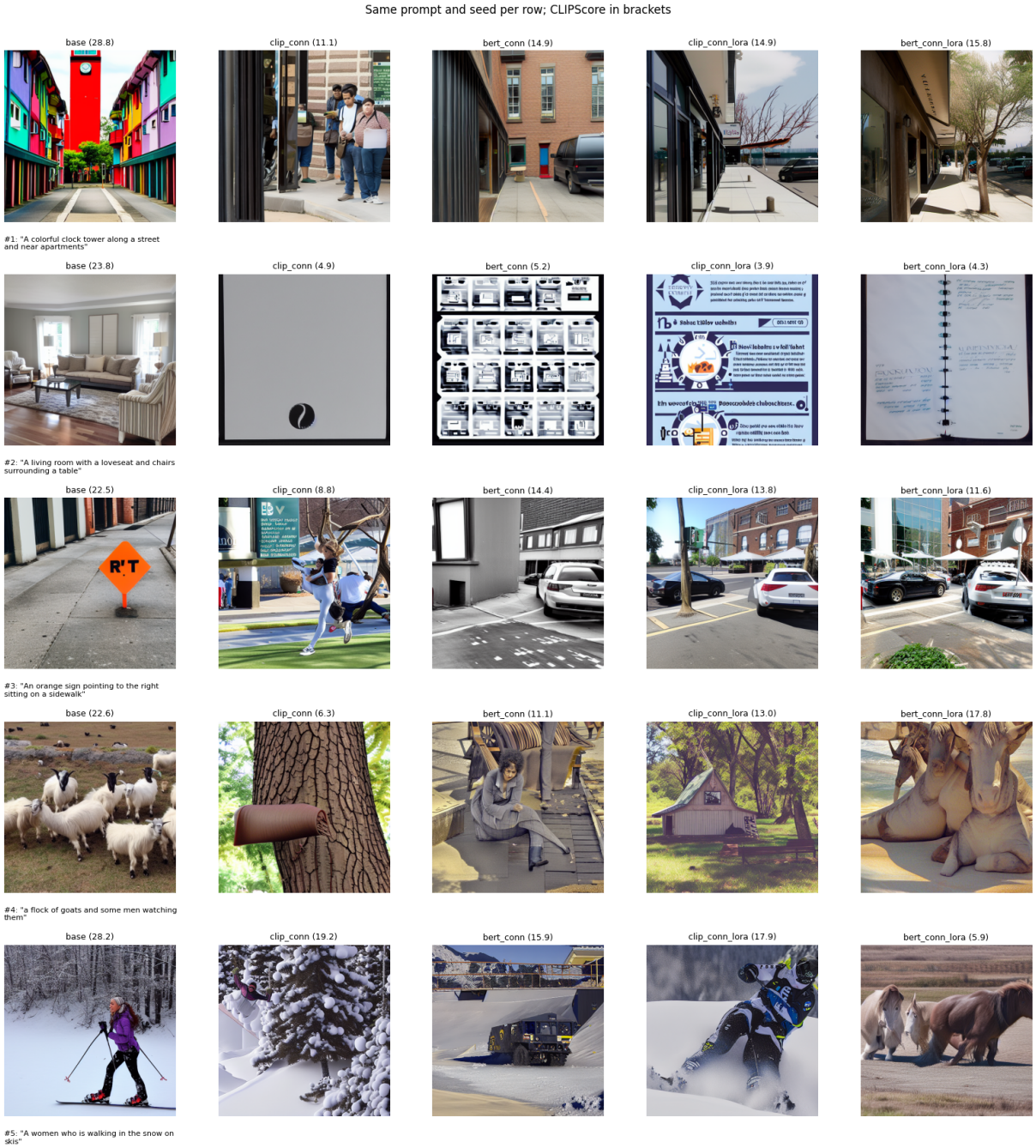



*Figure 4. Same prompt and seed per row. Base SD, then the four connector runs (CLIPScore in brackets).*

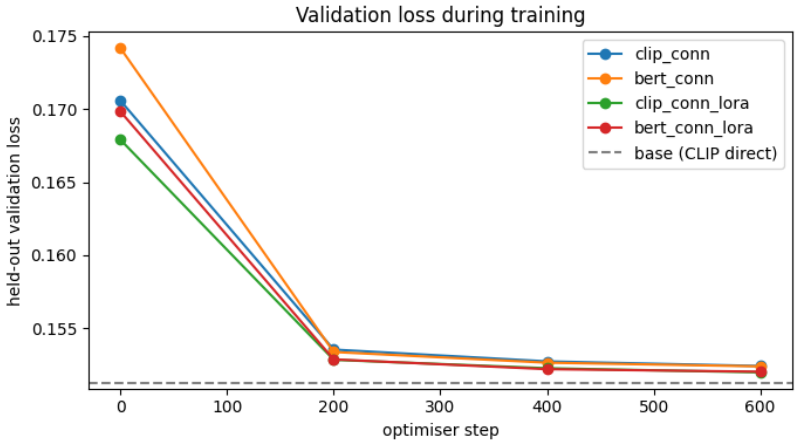



*Figure 5. Held-out validation loss during connector training; dashed line is the base model.*

## 8. What the results show

**BERT is worse than CLIP through the same connector.** The gap is about 2 points in both settings and both intervals stay below zero. This agrees with Praniti's no-training comparison, where BERT's sentence vectors related to CLIP's about as much as word counts do (CKA 0.53); here the same ranking shows up in the generated images.

**The bigger finding is that every connector run collapses.** All four score 11–14 points below the base model and beat it on almost no prompts. Figure 4 shows why: the images are realistic photos that only catch the rough scene (a street, snow, outdoors) and miss the subject (no clock tower, no living room, no goats, no orange sign). Even CLIP through a new connector loses most of what it had, so the cost comes from the connector being undertrained, not from either encoder. 600 steps on 3,000 images is far too little to teach a random-initialised bridge to speak the UNet's language.

**LoRA does not help yet.** Both LoRA comparisons include zero. With the connector this far off, a small UNet adapter has little to work with.

**Validation loss is a poor guide to prompt-following.** Across all runs it moves only between 0.1512 and 0.1524, while CLIPScore halves. Much of the denoising objective can be met from the noisy image alone, so it hardly notices whether the text is being used. CLIPScore and the images are the measures to trust here.

## 9. Limitations

• **Short runs.** 600 steps, under one pass over 3,000 images, for both parts. Part 2 in particular measures an undertrained connector, so it supports *BERT < CLIP at this budget*, not *BERT cannot work*.
• **One seed per run and 40 evaluation prompts.** The intervals cover prompt variation, not training-seed variation.
• **Checkpoint selection.** In Part 1 the reported checkpoint was chosen on the same 40 prompts it was scored on. The final checkpoint was also the best and scores rose at every checkpoint, so the bias is small, but a separate selection set is the clean fix (raised in Santhosh's comparison).
• **One automatic judge.** CLIPScore only; no FID/KID yet, and no object-level check of what is missing.
• **Environment.** Package versions were recorded but installs were not pinned; Part 2 ran in a session without Drive, so its weights live with that run's output folder.

## 10. What would make these results conclusive

Train long enough to converge (several passes) and pick checkpoints on a separate prompt set; score on the group's frozen eval_2000 split with FID/KID against Praniti's real-photo reference and her per-caption object lists; and repeat with at least three training seeds. For Part 2, a connector warm-start (first fitting it

to reproduce CLIP's features) would separate *how hard the bridge is to learn* from *how good the encoder is*.

## 11. Files and reproducibility

| File | Contents |
|---|---|
| milestone1_lora_finetuning.ipynb | Part 1 end to end: data, latent cache, LoRA training, evaluation, all outputs |
| results/lora_r8_pilot/ | eval_results.csv, summary.json, train/val logs, figures, 25 sample images, final LoRA weights |
| connector_bert_vs_clip.ipynb | Part 2 end to end, reusing Part 1's data cells so the splits are identical; all outputs |
| connector_experiment/ (run output) | results_table.csv, key_comparisons.csv, per-run summary.json, connector and LoRA weights, samples |

Environment: Tesla T4, PyTorch 2.11, Diffusers, Transformers, PEFT. Every random draw uses a fixed seed (data selection 42, validation noise 1234, generation seeds fixed per prompt).

## 12. Conclusion

Training a tiny fraction of Stable Diffusion works: LoRA on 0.185% of the UNet gives a small, statistically real gain in prompt-following in 14 minutes on a free GPU. Replacing the text encoder is much harder: through an identical connector BERT trails CLIP by about 2 CLIPScore points, but at this training length both connectors lose most of the prompt, so the encoder comparison is real yet small next to the cost of learning the bridge at all. The next step for both is the same: longer training, a separate selection set, and image-quality metrics on the group's frozen splits.